SE-JEPA: Self-Evolving Latent World Models

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Abstract

Latent world models let agents plan by imagining the outcomes of their actions, but they stop learning once deployed: when objects or appearances shift, prediction errors become planning failures. Test-time adaptation such as AdaJEPA repairs these errors within an episode but then resets, meeting every recurring shift as if for the first time. We therefore seek a world model that remembers what it learns. However, we find that simply keeping the adapted weights fails: updates across episodes are nearly orthogonal, and accumulating them degrades planning. To overcome this, we present SE-JEPA, a self-evolving latent world model. Inspired by complementary learning systems, it pairs fast within-episode adaptation with slow cross-episode consolidation, replaying experience into slow learners, one per self-discovered domain. Supervised only by its own prediction errors, SE-JEPA decides what to consolidate, where to store it, and when to trust it for planning. Because its pretrained model is never overwritten, nothing it already knows can be forgotten. It requires no shift labels or task boundaries and benefits both gradient-based and sampling-based planners. Across unseen object shapes, visual shifts, and deformable rope and granular manipulation, SE-JEPA grows stronger the longer it is deployed, setting a new state of the art by a wide margin.

1. Introduction

Visual world models let an agent plan a push by predicting where the object will end up and comparing that prediction with an image of the goal [22, 36, 42]. An unfamiliar shape can turn differently under the same push. A change in color can alter how the model encodes a familiar motion. In both cases the model gets feedback as soon as the agent acts. The discrepancy between the predicted and observed motion gives it a chance to improve the next plan.

AdaJEPA [35] uses this feedback to adapt the model during an episode. It resets the adapted parameters when the episode ends. If the same shape returns with a different target, the agent has to learn how it moves again. Keeping every parameter update also carries over adjustments specific to the preceding trajectory. On a stream of recurring shapes, this rule solves fewer goals than resetting between episodes (Sec. 3.1).

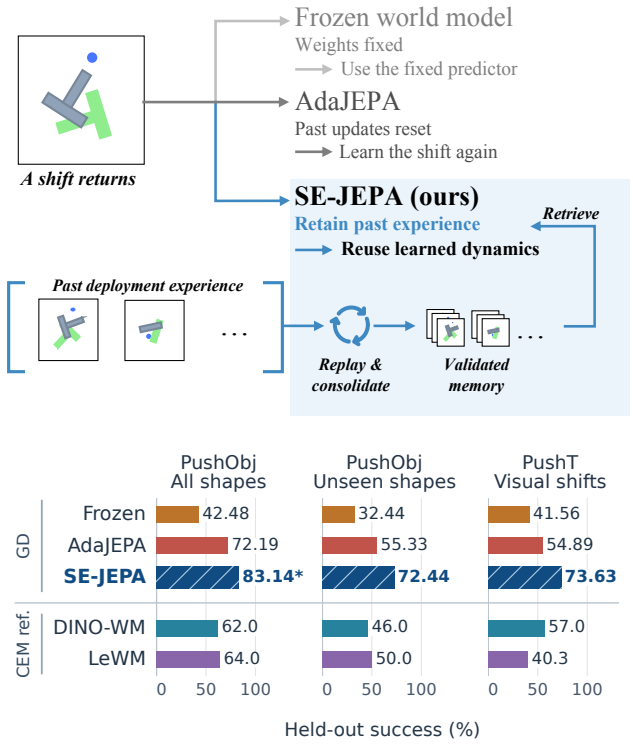


Figure 1. **Planning when a shift returns.** The schematic shows how SE-JEPA retrieves a predictor learned from earlier encounters. Bars show held-out success with GD and, in a separate group, CEM results for DINO-WM and LeWM. The star marks SE-JEPA with both familiarity protections disabled (Sec. 4.5).

A trajectory contains information that survives a change of goal. It records which motion actually followed each push. Several encounters with the same shape can therefore teach a predictor how that shape moves across a wider range of starts and goals. The challenge is to recognize which encounters belong together when neither the shape nor the onset of a shift is labeled.

SE-JEPA combines fast adaptation during an episode with slow learning across encounters, inspired by complementary learning systems [23]. It keeps the pretrained model available to the planner and uses that model's errors

to decide which trajectories to retain (Fig. 1, top). It groups trajectories by the direction of their fast updates and by how well one predictor explains their transitions. Each group trains a separate predictor by replay. Before updating that predictor on a new trajectory, SE-JEPA tests how well it predicts the observed transitions. Predictors that consistently outperform the pretrained model on these forecasts become candidates for planning.

We test what this memory retains by changing the goal after deployment. For instance, an I-shaped object can reappear in a new pose with a different target. The agent must choose a new sequence of pushes, but what it learned about how the object rotates can still guide the search. We fix the learned memory and give the planner fresh targets. It can read a predictor and adapt within the episode as feedback arrives.

With gradient descent (GD), SE-JEPA reaches more held-out goals than episodic adaptation across unseen shapes and visual shifts (Fig. 1, bottom). The gains are especially large when the object or its goal changes color. They also hold with the cross-entropy method (CEM). On both rigid-object tasks, SE-JEPA reaches more goals by the midpoint of the CEM replan budget than AdaJEPA reaches by the end (Fig. 5).

These tasks use ResNet world models trained on manipulation trajectories. Replay also improves a DINO-WM predictor for rope and granular manipulation, bringing the final configuration closer to the goal in both tasks. A rope predictor learned in one pass transfers to previously unseen instances, showing that the experience gained on one set of configurations can guide action on another.

2. Related Work

World models for control. World models support control in several ways. Embed to Control [37], PETS [6], and PlaNet [9] predict the consequences of candidate actions for planning. World Models [8], SimPLe [16], IRIS [24], and the Dreamer family [10–12] use learned simulators to train a policy. MuZero and EfficientZero [29, 38] learn models for tree search, while TD-MPC and TD-MPC2 [13, 14] combine latent dynamics with value learning. We focus on goal-image planning, where a model can learn directly from the discrepancy between a predicted and an observed transition.

Predictive representations and JEPA. Predicting future representations is central to JEPA [18] and earlier work on predictive embeddings [30, 32]. Self-supervised objectives learn visual features by matching related views or regularizing the embedding distribution [3, 7, 40]. I-JEPA [1] predicts embeddings of masked image regions, V-JEPA [4] extends this idea to video, and V-JEPA 2 [2] conditions predictions on actions for robotic planning. For control, the encoder can draw on pretrained visual or manipulation features [5, 25, 26]. DINO-WM [42] predicts frozen patch embeddings, whereas LeWorldModel (LeWM) [22] trains its encoder and predictor together from pixels. Temporal straightening [36] shapes the latent trajectories used for planning, and DDP-WM [39] separates primary dynamics from background updates. Our method addresses how to update the predictor once the agent begins to act.

Test-time adaptation. Test-time training and entropy minimization update deployed models from unlabeled observations [31, 33], including under changing distributions [34]. In control, each executed action supplies a target for the dynamics predictor. AdaJEPA [35] uses this signal within an episode and resets the weights at its end. JEPA-TTT [41] studies a dynamics shift that persists across episodes. It trains one predictor by replaying overlapping windows from the accumulated interactions. SE-JEPA addresses recurring shifts. It keeps the pretrained reference, groups compatible trajectories, and chooses among predictors that have improved on the reference on fresh trajectories.

Continual learning and complementary memory. Complementary learning systems pair rapid acquisition with slow consolidation [23]. Continual learners replay earlier experience [27], constrain updates to protect past tasks [17, 21], preserve earlier outputs through distillation [19], or retain separate task modules [28]. For a planner, retrieving a predictor also changes the actions taken and hence the data collected next. SE-JEPA uses the first observed transitions of each episode to select among eligible predictors, including the pretrained reference.

3. SE-JEPA

3.1. Motivation

Consider two episodes in which the agent pushes the same unfamiliar object toward different goals. The first may reveal how an off-center push rotates the object. That observation can help the agent choose a push in the second episode, even when it approaches the object from a different side. An episodic learner corrects its predictions along the first trajectory [35]. Carrying its weights forward assumes that the next trajectory calls for a similar correction.

We compared these choices on the same stream of 100 PushObj goals. Inheriting the previous episode's fast weights recovered 11 goals that a reset had missed, but lost 25 that the reset had solved (Appendix D). Consecutive weight updates also point in nearly orthogonal directions (Appendix D). A trajectory offers a different way to retain what the agent learned. Even a push that rotated the object away from its goal tells the agent which rotation followed

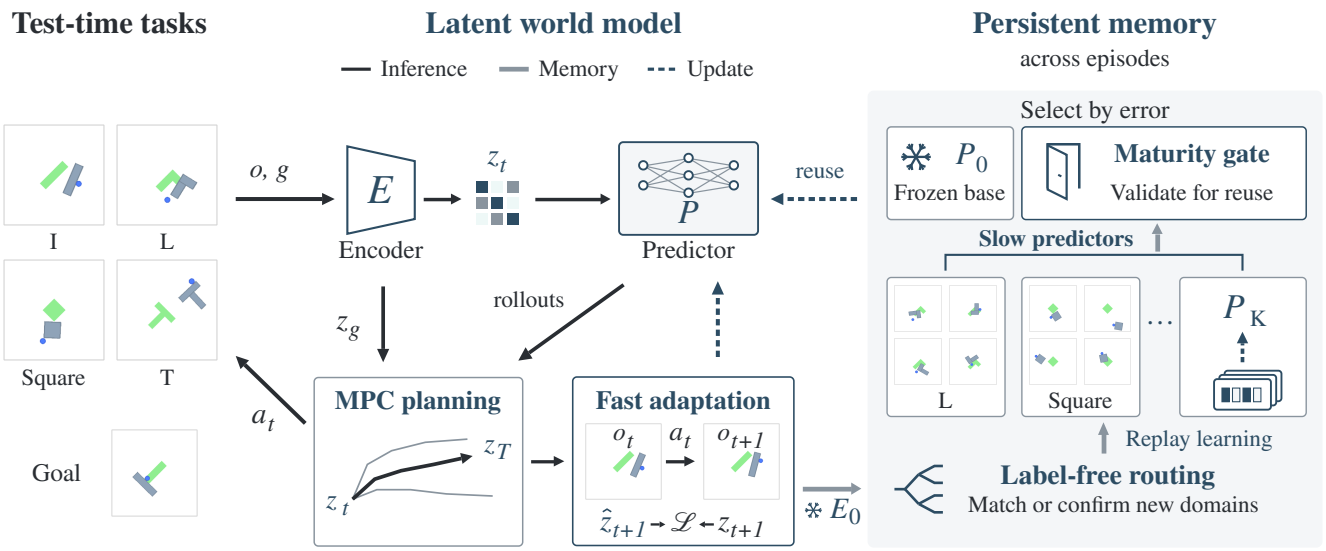


Figure 2. **SE-JEPA architecture.** The fast loop plans in latent space and adapts the predictor from observed transitions. At the end of an episode, the slow system re-encodes the trajectory with the frozen encoder, assigns unfamiliar experience to a self-discovered domain, and trains that domain's slow predictor by replay. Predictors that pass the maturity gate compete with the frozen base at the start of the next episode. Fast parameters reset at every episode; slow predictors and replay persist. Shape names illustrate memory contents and are not supervision.

that contact. If a later goal calls for that rotation, the planner benefits from having a predictor that can anticipate it. The learning signal is available as soon as the next observation arrives, regardless of whether the episode eventually succeeds.

SE-JEPA keeps a fast model for the current episode and trains a slow predictor on trajectories from multiple encounters (Fig. 2). The agent receives no labels telling it when a shape or appearance recurs. It groups trajectories by how they change the model and how well a common predictor explains them. When a new goal arrives, the first few actions provide evidence for which predictor to use.

3.2. Fast adaptation and replay consolidation

Each episode begins with the pretrained encoder E_0 and predictor P_0. To plan toward a goal image, the agent encodes the current observation and the goal, rolls out candidate action sequences with the current predictor, and chooses the sequence with the lowest latent goal cost [22, 42]. It executes the first action block. Following AdaJEPA [35], it then takes one Adam step on the prediction loss over recent transitions to update the predictor's last block and the encoder head before replanning. These fast changes last for the episode. During the first few replans, the agent can also retrieve a slow predictor as a base for adaptation (Sec. 3.4).

Across episodes, SE-JEPA maintains a replay buffer $\mathcal{R}_k$ and a slow predictor P_{ϕ_k} for each group k of compatible trajectories, which we call a *predictive domain*. After an episode, the frozen encoder E_0 re-encodes the executed trajectory. This puts observations from different episodes in a common latent space even as the fast encoder changes. Once a trajectory is assigned to a domain, the agent adds it to $\mathcal{R}_k$. The slow predictor trains on windows sampled from that buffer by minimizing

$$\mathcal{L}_k(\phi_k) = \mathbb{E}_{(x,z^+)\sim\mathcal{R}_k} \left\| P_{\phi_k}(x) - z^+ \right\|_2^2. \quad (1)$$

Here x contains a short history of observation embeddings and executed actions, and z^+ is the embedding of the next observation. Each slow predictor starts as a copy of P_0. A replay update samples episodes from the buffer and draws a short window from each. The same update can therefore include a push that translated an object and an earlier push that turned it. Both windows describe observed motion in the coordinates of E_0, even if fast adaptation changed the encoder differently in the two episodes. The goal image enters when the planner uses this predictor to compare candidate futures.

3.3. Discovering predictive domains

The frozen reference provides a first test of whether an episode contributes new experience. We take the geometric mean of its next-frame prediction errors over the first few observed transitions. When the error falls below a calibrated threshold, the agent uses P_0 as the base for fast adaptation and skips the direct memory write for that episode. This reserves new replay entries for episodes the reference predicts poorly.

For an unfamiliar episode e, let Δ_e denote the fast parameter change. Its direction $u_e = \Delta_e / \|\Delta_e\|_2$ helps identify episodes that call for similar corrections. If the agent retrieved a slow predictor, its domain is the first candidate for the new trajectory. Otherwise, the agent ranks stored domains by the cosine similarity between u_e and each domain's prototype. A candidate accepts the trajectory if its predictor passes the error check in Sec. 3.4. Unassigned episodes wait in a pool. When a new direction aligns with those of two waiting episodes, the three trajectories establish a new domain.

Two contacts with the same object can induce different fast updates. Delaying domain creation gives the planner a way to keep an unusual contact until it sees a similar one again. The pool retains the founding trajectories, so learning can begin with examples from several encounters. As more evidence arrives, two domains merge if each predictor transfers to the other's recent episodes and both outperform the reference. Testing transfer in both directions checks that the proposed merger covers the experience of each group. A mature domain can also absorb a smaller group whose trajectories it already predicts well. Appendix A gives the thresholds and the rules for maintaining each prototype.

3.4. Validated retrieval for planning

A slow predictor earns a place in the planner by predicting trajectories it has yet to train on. Before adding trajectory τ_e to a buffer, the agent rolls out the candidate predictor and the pretrained reference under the actions actually taken. Their open-loop errors in the frozen latent space give the ratio

$$r_{e,k} = \frac{\mathcal{L}_{\text{roll}}(\phi_k; \tau_e)}{\mathcal{L}_{\text{roll}}(\theta_0; \tau_e)}. \tag{2}$$

Here θ_0 denotes the weights of P_0. Comparing both predictors on the same sequence of pushes makes the reference specific to the encounter. A large error on a difficult turn, for example, can still favor the slow predictor if it forecasts the turn more accurately than P_0. A large ratio rejects a write. Accepted episodes leave a record of how well the predictor transferred to a fresh trajectory, since the agent scores each one before replaying it. A domain becomes eligible for retrieval once this record shows a sustained improvement over the reference (Appendix A).

During the first three replans, the agent compares eligible predictors and P_0 on the transitions observed so far, $\mathcal{B}_{e,t}$. It selects the predictor with the lowest recent error,

$$k^\star = \underset{k \in \{0\} \cup \mathcal{K}_{\text{eligible}}}{\arg\min} \mathcal{L}_{\text{pred}}(\phi_k; \mathcal{B}_{e,t}), \qquad \phi_0 = \theta_0. \tag{3}$$

The selected weights become the base of the fast predictor, preserving the corrections already made in the current episode. For an episode the reference recognizes as familiar, the base stays at P_0. Thus past trajectories determine which predictors are eligible, and the current actions and observations determine which one guides planning.

4. Experiments

4.1. Evaluation setup

Tasks and backbones. Each episode starts with a scene and an image goal. The planner observes the consequences of its actions and receives no shift labels or task-change signals. We study shape and appearance shifts with ResNet-based world models trained on task trajectories [15]. In *PushObj*, pretraining covers T, L, Z, and plus shapes. Deployment also introduces I, square, and small tee. *PushT* keeps the T shape and changes its appearance through blur, noise, darkness, or recoloring of the agent, object, or goal. We also test replay in two deformable manipulation tasks, *rope* and *granular*, using DINO-WM with pretrained DINOv2 features [20, 26, 42].

Deployment and evaluation. For the standard PushObj and PushT comparisons, each planner runs three learning chains. Targets are resampled over five rounds, with 50 episodes per shape or visual condition in each round. We then fix each learned memory and test it on held-out goals from three new seeds. Retrieval and fast adaptation stay active. We report the fraction of goals reached and the mean number of replans, counting the full budget for failures. On rope and granular, we isolate slow consolidation by training one predictor per task with fast adaptation disabled. The planner revisits the same 50 instances over four passes. We measure Chamfer distance (CD) between the final and goal configurations and test the rope predictor separately on new instances.

Baselines and planners. Frozen (pretrained) plans with a fixed model. AdaJEPA [35] adapts within each episode. On rigid objects these two baselines use released task-specific checkpoints and the same held-out targets as SE-JEPA. We evaluate both GD and CEM, executing five actions per replan up to a budget of 30 replans. Additional CEM baselines include DINO-WM, LeWM, DDP-WM, and planners using DINOv2 CLS or R3M features [22, 25, 26, 39, 42]. For PushT, we use DreamerV3 [12] and IRIS [24] as offline-trained predictors for CEM. The published two-shape PushObj results of Zhou et al. [42] are listed separately in Appendix C. Appendix B and Appendix C give the target sets, checkpoints, and planning protocols.

4.2. Planning with memory on held-out targets

Which goals benefit from memory? On shapes absent from pretraining, held-out CEM success is about 15 percentage points higher with SE-JEPA than with AdaJEPA (Table 1). Figure 7 shows how the gain varies by shape

Table 1. **Goal-conditioned planning with CEM.** PushObj covers seven shapes, three unseen during pretraining. PushT covers six visual shifts, including three recolorings. Success rates are percentages. Steps count MPC replans, with failures assigned 30. Italic Chamfer distances (CD) come from Zhou et al. [42]. $^{\circ}$ marks offline PushT adapters and ^{T} marks T-only checkpoints. † marks PushObj All with familiarity checks disabled for reads and writes and maturity screening retained. The other SE-JEPA entries use the standard task configurations in Appendix B.

	PushObj			PushT visual shifts			Rope	Granular
Method	All (7) ↑	Unseen (3) ↑	Steps ↓	Shift (6) ↑	Color (3) ↑	Steps ↓	CD ↓	CD ↓
DreamerV3° [12]	—	—	—	5.7	0.0	28.91	*2.49*	*1.05*
IRIS$^{\circ}$ [24]	—	—	—	6.0	6.7	28.85	*1.11*	*0.37*
R3M^{T} [25]	18.9	18.0	25.49	21.7	27.3	25.00	*1.13*	*0.95*
DINO CLS^{T}	14.9	10.0	26.77	34.7	35.3	22.38	*0.84*	*0.79*
DINO-WM [42]	62.0	46.0	16.15	57.0	65.3	17.32	0.491	0.301
LeWM [22]	64.0	50.0	16.28	40.3	16.0	21.45	0.786	0.322
DDP-WM [39]	57.7	43.3	17.59	52.7	57.3	18.39	0.677	0.259
Frozen (pretrained)	43.8	36.2	21.08	48.3	33.6	20.10	0.491	0.301
AdaJEPA [35]	70.4	54.4	16.02	60.2	42.2	18.46	0.629	0.304
SE-JEPA	**79.62**†	**69.9**	**13.53**	**76.1**	**65.2**	**14.46**	**0.252**	**0.220**

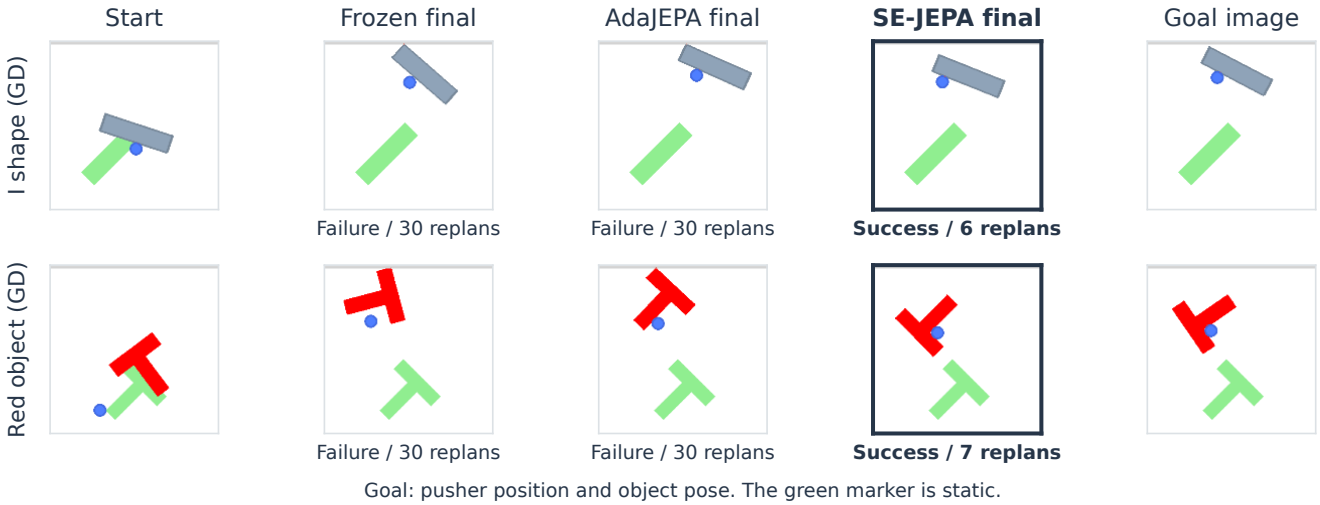


Figure 3. **Persistent memory resolves failures of episodic adaptation.** Selected GD examples on an unseen I shape (top) and red-object shift (bottom). Frozen and AdaJEPA exhaust 30 replans in both cases; **SE-JEPA** succeeds in six and seven, respectively. Each row shares the same start and goal. The goal image specifies the pusher position and object pose; the green marker is static.

and planner. Among the three unseen shapes, both planners gain most on the I shape, followed by the square and the small tee (Table 6). Each shape recurs during deployment, but the agent must bring it to a new pose at test time. What transfers across these goals is the predictor's experience of how pushes move the object. The planning objective still comes from the new goal image.

PushT holds the object geometry fixed and changes how the scene looks. Recoloring the object provides a particularly clear comparison. With CEM, the frozen and episodically adapted models reach the same fraction of red-object goals. With memory, success rises by over thirty percentage points relative to either baseline (Table 7). The red-goal condition also shows a large gain. In both conditions the agent can draw on pushes observed in earlier scenes with that appearance, even though the current episode supplies a different target.

How much interaction does planning require? Figure 5 follows the full model as the replan budget increases. Halfway through the budget, SE-JEPA has already solved a larger fraction of goals than AdaJEPA reaches with the full budget. This holds on the seven PushObj shapes and on the seven PushT visual conditions, both with CEM. The curves count every goal presented to each method, including those still unsolved when the budget expires. Earlier successes

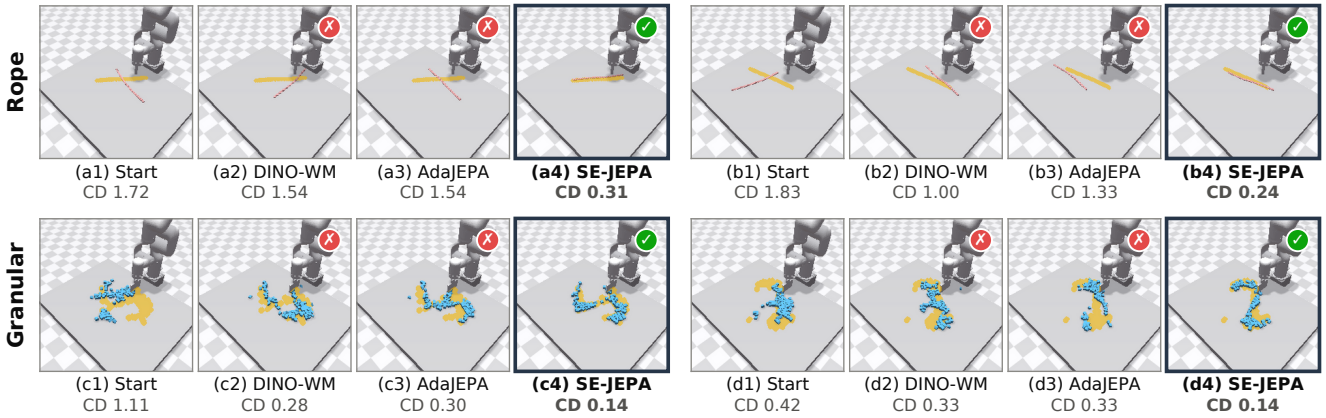


Figure 4. **Consolidated predictions support deformable manipulation.** Real FleX simulator trajectories for rope (top) and granular material (bottom), with two instances per task. Each group shows the start and the three methods' final states after 15 pushes; gold denotes the goal. These third-pass examples illustrate successful slow-only consolidation and were selected where both baselines fail. Frozen DINO-WM frames replay its executed actions. The aggregate fourth-pass results are reported in Table 1.

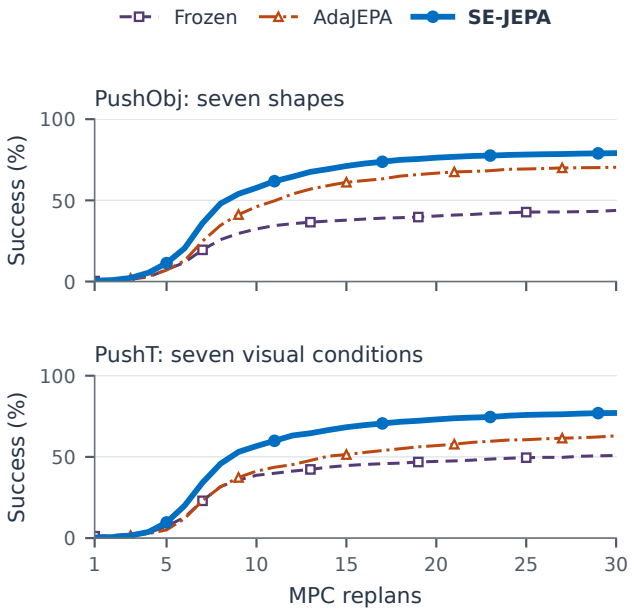


Figure 5. **Higher success within the same interaction budget.** Cumulative held-out success under CEM. Curves use all seven shapes (top) and all seven visual conditions (bottom).

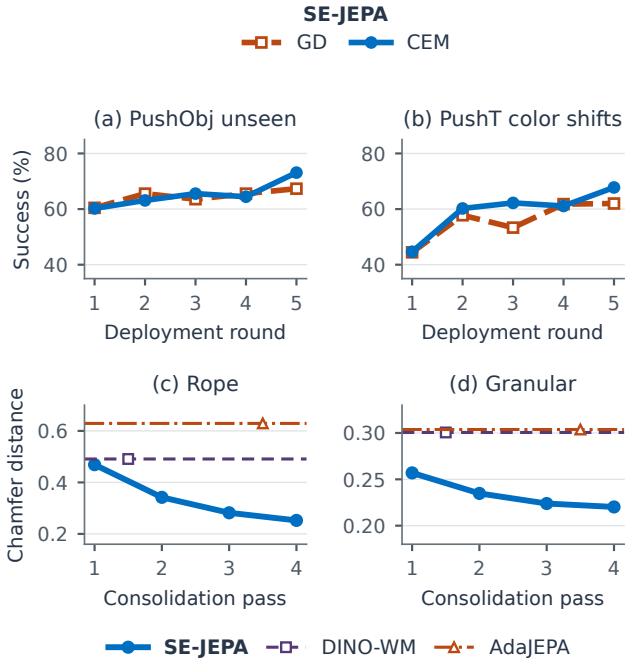


Figure 6. **Deployment experience improves two backbone families.** Top: ResNet-based models, with five rounds of online success on unseen PushObj shapes and three PushT color shifts; curves average three learning chains with targets resampled each round. Bottom: DINO-WM, with four consolidation passes over 50 fixed rope or granular instances; curves show mean CD. The horizontal lines are frozen DINO-WM and AdaJEPA.

matter when the controller has a fixed number of pushes to reach the goal. With GD, the same ordering appears in the mean number of replans on both tasks (Appendix B).

Consolidation on a second backbone. On rope and granular, replacing the frozen DINO-WM predictor with the consolidated predictor brings both materials closer to their target configurations (Table 1). The local deformable evaluation gives each planner the same budget of 15 pushes. A lower final CD means the planner uses those pushes to better match the rope's shape or the distribution of particles in the goal. The planner has access to one slow predictor per task and makes no fast updates during an episode. The im-

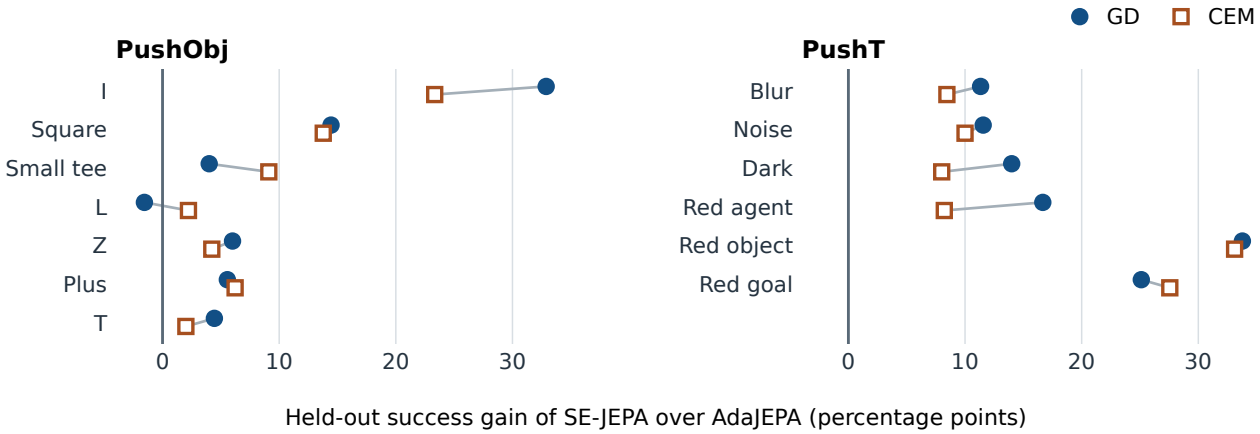


Figure 7. **Where memory improves planning.** Each point is the difference in held-out success between full SE-JEPA and AdaJEPA for one planner, in percentage points. Positive values favor SE-JEPA. I, square and small tee are the three shapes first encountered during deployment.

provement therefore tests the part of SE-JEPA that learns from previous trajectories, using DINOv2 features in place of the ResNet features used on rigid objects.

4.3. Qualitative results

Figure 3 shows two episodes in which all three methods start from the same scene and aim for the same goal. On the unseen I shape, SE-JEPA reaches a pose that both baselines miss. In the red-object episode, the baselines leave the T at the wrong orientation, while SE-JEPA turns it into the goal pose. Figure 4 shows the final rope and particle configurations after the executed pushes. With the slow predictor alone, the planner moves the rope toward its target shape and the particles toward the goal distribution. Appendix E gives additional trajectories.

4.4. Learning over deployment

When targets are resampled each round, both GD and CEM finish deployment with higher online success on unseen shapes and recolored scenes than in the first round (Fig. 6, top). These curves track performance as memory grows. Table 1 tests that memory after learning, on independent rigid-object goals.

Revisiting the same rope and granular instances reduces final CD over successive passes (Fig. 6, bottom). To test whether the rope predictor also transfers, we froze it after its first pass and used it on 91 previously unseen instances. On those instances, mean CD was 47% lower than with the pretrained DINO-WM predictor (Appendix B). The two evaluations ask different questions about the same learned dynamics. On the repeated stream, changes in final CD show how another pass changes control on the same set of tasks. On the fresh rope instances, both planners start from the same new configurations. The one-pass predictor guides the rope closer to these new goals with its weights fixed throughout the test.

Which encounters use memory? Across seven PushObj rounds, about one episode in four adds experience to a replay buffer, and a similar fraction finishes with a slow predictor. This holds for both planners (Appendix D). An episode can contribute training data to a domain whose predictor is still awaiting validation. The pretrained model remains available on familiar encounters. On the four shapes seen in pretraining, SE-JEPA also improves held-out success over AdaJEPA with both planners (Appendix D).

4.5. Ablation study

Reading and writing on familiar episodes. When the pretrained model predicts an episode well, SE-JEPA normally uses it for planning and skips the direct replay write. We remove the read and write protections separately and together (Table 2). Removing the read protection lets mature slow predictors compete for retrieval. Removing the write protection lets familiar trajectories enter the usual replay selection process. All four settings use the same maturity criterion.

Table 2. Effect of familiarity protection on PushObj success (%). Each run uses the same five-round learning stream and 1,050 held-out goals across seven shapes. Dashes indicate configurations with no reported test.

Familiarity protection	GD	CEM
Full SE-JEPA	81.43	78.57
Disable read protection only	76.95	—
Disable write protection only	79.43	—
Disable both protections	83.14	79.62

With GD, removing either protection alone lowers success on this stream. Removing both gives the highest GD success among the four settings. With CEM, removing both gives a rate close to the full model's.

What should carry across episodes? Table 3 compares ways of retaining experience on the same seven-shape stream. All variants start from the same backbone weights, and each learned state faces the same held-out targets. Reusing the previous episode's final weights solves fewer goals than resetting to the shared checkpoint. Averaging the fast weights across episodes recovers some of those goals. Training a slow predictor on stored transitions gives a larger gain, even when every shape shares that predictor.

Table 3. What carries across goals? All variants start from the same pretrained weights and use GD. Success and mean replans cover the same 1,050 held-out PushObj goals.

Update between episodes	Success (%)	Replans
Full SE-JEPA (10^{-4})	81.43	11.37
AdaJEPA, same backbone weights	76.95	12.54
Inherit fast weights	64.67	16.12
Shared slow predictor (EMA)	75.71	12.84
Shared slow predictor (replay)	81.24	11.41
Without maturity screening	80.86	11.46

The difference is most visible on the I shape. With a single slow predictor in each case, replay solves 44 more of the 150 I-shape goals than weight averaging. On the four pretraining shapes, the two totals differ by just one goal. This comparison isolates a practical advantage of retaining trajectories. Replay can revisit the contacts made on one shape after the agent has moved on to another, fitting the predictor to their observed consequences. On this stream the shared replay predictor nearly matches the predictors trained separately for each discovered domain. Appendix D gives the slow learning rate sweep.

5. Conclusion

SE-JEPA lets a visual planner reuse what it learned on earlier encounters. It keeps trajectories that the pretrained predictor struggles to explain, groups compatible experience for replay, and tests the resulting predictors on fresh trajectories before using them for control. With task-trained ResNet models, this memory improves held-out planning success and reduces replans on rigid objects under both GD and CEM. With a DINO-WM backbone, slow consolidation lowers goal error for rope and granular manipulation, and a rope predictor trained in one pass transfers to new instances.

Future work. Two objects can respond alike to an initial push and diverge at the next contact. A longer interaction history, or a short probe chosen to distinguish candidate predictors, could help route these episodes. We also observed that, late in deployment, prediction error can keep falling after planning success levels off. One way to study this gap is to test whether a more accurate predictor ranks candidate pushes in a better order near the goal. That test could guide the choice of planning horizon and how much search to spend on the final contacts.

LLM Usage Statement

Large language models (OpenAI Codex and Anthropic Claude) were used to draft and revise substantial portions of the manuscript text, and to assist with literature and source checks, LaTeX organization, figure generation, and result aggregation. They also assisted with experimental code, including offline world-model adapters and experiment orchestration. All quantitative results are taken from saved experimental outputs or explicitly cited published sources; language-model output is not used as experimental evidence. The authors take full responsibility for the claims, citations, code, and final verification of the manuscript.

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Supplementary Material

A. Implementation and Memory Management

Compute environment. Model training and evaluation were conducted on systems equipped with 8× NVIDIA GeForce RTX 5090 GPUs and 8× NVIDIA A100 GPUs.

A.1. Fast and slow learning

The label-free rigid-object configuration builds on the episodic adaptation loop of AdaJEPA [35]. At every replan, the most recent observed transitions provide a self-supervised next-latent loss. The predictor's last block and the encoder's designated projection/head parameters are adapted with one Adam step. The episode-local predictor and encoder changes are reset at the next episode. A retrieved slow predictor supplies the predictor initialization; it does not replace the frozen reference encoder.

Persistent replay is re-encoded by the reference encoder. Each slow learner is a full predictor copy, initialized from the pretrained predictor rather than from the final fast parameters of its founding episode. For each slow optimization step, up to eight stored episodes are sampled with replacement; each contributes a random window of up to six latent frames. The prediction objective averages one-step observation-feature errors over the available history windows. Action features condition prediction but are not reconstruction targets. Standard writes use 15 Adam steps. Founding a learner after n stored episodes uses $15n$ steps to process the accumulated evidence. Merging likewise adds replay updates for absorbed experience.

Replay stores compact latent trajectories rather than images. The pending pool is bounded, and merging and absorption remove redundant predictor copies as compatible domains are recognized.

A.2. Order of decisions

Algorithm 1 summarizes a learning episode. Familiarity uses the geometric mean of newest-frame prediction errors on up to the first three observed transitions. The threshold is calibrated once on the shape backbone and transfers unchanged to the visual experiments.

The selected predictor's error is evaluated before it learns from the current episode, so maturity reflects transfer to data the predictor has not fitted. In read-only evaluation, the episode-end write, validation-history update, slow training, merging, and absorption steps are all disabled.

A.3. Fingerprints and delayed domain founding

A fingerprint concatenates the adapted predictor and encoder-head changes and is normalized to unit length. Domain prototypes accumulate normalized fingerprints from episodes planned from the reference; episodes starting from a slow predictor do not redefine that reference-based prototype. If an episode has already read a learner, it preferentially writes to that learner's domain unless the prospective error ratio rejects the write. Otherwise, it tries compatible domain directions in descending cosine order.

An unassigned episode enters a pending pool. It confirms a new domain when at least two pending fingerprints align with its own above the new-domain threshold. These members need not be pairwise aligned with one another; the incoming fingerprint is the common reference for this confirmation. The members and the new episode are transferred together to the new replay. No ground-truth shape or appearance category participates in this procedure.

Table 4. Label-free configuration used for PushObj and PushT. The seven-round stream in Appendix D uses a slow learning rate of 2×10^{-4}.

Setting	Value
Fast predictor learning rate	5×10^{-4}
Fast encoder-head learning rate	10^{-5}
Fast steps per replan	1
Recent adaptation buffer	5 transitions
Slow learning rate	10^{-4}
Slow steps per standard write	15
Slow batch size (episodes)	8
Slow sampled window (frames)	up to 6
Familiarity threshold τ_f	0.0146
Existing-domain cosine threshold	0.10
New-domain cosine threshold	0.05
Episodes to confirm a domain	3
Pending-pool capacity	40
Minimum episodes to train a learner	3
Retrieval window	first 3 replans
Maturity history length m	3
Maturity error-ratio threshold τ_m	0.9
Write rejection ratio	1.2
Merge evidence per direction	2 episodes
Merge relative tolerance	0.5
Fragment absorption interval	25 episodes
Absorption error-ratio threshold	0.8
Fragment-size limit	< 6 episodes

A.4. Merging and fragment absorption

For a new unfamiliar episode, every existing trained learner is evaluated before training. The resulting ratios provide cross-domain evidence. Let $G_{A\to B}$ be the geometric mean error ratio of learner B on recent episodes assigned to A, and $G_{A\to A}$ the corresponding ratio for learner A. Domains merge only if both directions satisfy

$$G_{A\to B} \le \min\{1, (1+\epsilon_m)G_{A\to A}\}, \tag{4}$$

Algorithm 1 One deployment-learning episode of SE-JEPA

Require: Frozen (E_0, P_0), memory $\mathcal{M}$, initial observation o, goal g
Reset fast parameters to (E_0, P_0)
Initialize recent buffer $\mathcal{B}$ and episode trajectory τ
while budget remains and task is not complete **do**
    Optimize an action sequence with the current fast model
    Execute its first block; observe the next state
    Append real transitions to $\mathcal{B}$ and τ
    if within the early retrieval window **then**
        Form candidates: P_0 and mature slow predictors
        Restrict to P_0 if the observed episode is familiar
        Select the smallest observed prediction error
        Load the selected predictor, retaining the fast offset
    end if
    Adapt the fast model on recent real transitions
end while
Encode τ with frozen E_0
Measure pre-update rollout errors of existing predictors
if episode is unfamiliar **then**
    Propose a domain using retrieval or an update fingerprint
    Reject an incompatible write, or retain pending evidence
    if a domain is assigned or newly confirmed **then**
        Record prospective validation when a learner exists
        Append latent experience; update its slow predictor
        Check bidirectional merges and periodic absorption
    end if
end if
Discard episode-local fast parameters

with the symmetric condition for $B \rightarrow A$. Each direction requires two jointly evaluated episodes in the primary configuration. The larger domain retains its learner and incorporates the union of stored experience. Merged identifiers are not reused.

Every 25 episodes, small immature domains can be absorbed into a mature predictor that explains their experience with an error ratio at most 0.8. The primary configuration limits absorption to fragments with fewer than six episodes and fewer than three maturity measurements. Pending-pool absorption and the additional absolute-error gate are disabled in the shape and visual experiments.

B. Evaluation Protocols

B.1. Shape shifts

The shape learning stream has five deployment rounds over seven shapes, with 50 episodes per shape in every round. The held-out benchmark uses target seeds 2000, 2100, and 2200, each with 350 episodes. I, square, and small tee are unseen in pretraining; T, L, Z, and plus are seen.

For the paired planner comparison in Table 5, each planner has three learned memory states. Every state is evaluated on all three held-out seeds. For each seed we average over episodes and states, then compute the standard deviation across seed-level means. The CEM states are S7t3, S7t3b, and S7t3s700, where S7t3b reruns S7t3 after a run in which no learner matured. These states supply the SE-JEPA Unseen and Steps entries for PushObj in Table 1. Frozen and AdaJEPA use the same three held-out seeds. We use seed 2000 for the external PushObj baselines.

The SE-JEPA PushObj All results marked in Table 1 and shown in Fig. 1 come from a separate ablation that disables familiarity checks on memory reads and writes and retains maturity screening. We train one chain per planner and evaluate its final state on 1,050 episodes across the three held-out seeds.

B.2. Visual shifts

The paired visual evaluation contains the original T condition, blur, salt-and-pepper noise, darkness, red agent, red object, and red goal. Three five-round learning chains resample targets every round, with 350 episodes per round. Held-out testing uses three target seeds and 50 episodes per condition. Direct checks of source-episode identifiers confirm that held-out sources never occur in the learning streams and that the three test seeds use disjoint sources.

The color-learning curve in Fig. 6 aggregates red-agent, red-object, and red-goal episodes. For each round and learning chain, it sums successes over the three conditions and divides by 150 episodes, and plots the mean of the three chain-level rates. The archived statistics include their sample standard deviation. The three conditions share the same 50 source targets within a chain and round.

The external visual baselines use the established VIS50 collection with planning seed 100. The external comparisons use its six shifted conditions, with 50 paired initial/goal states per condition.

B.3. Success and interaction counts

For rigid objects, GD uses 100 optimization steps; CEM uses 10 iterations with 200 samples and 30 elites per iteration. Rigid-object success requires a joint Euclidean error below 20 simulator units across the pusher and object positions, and an object orientation error below $\pi/9$. Each executed model action contains five physical actions, and

an episode has at most 30 MPC replans. For episode i with success indicator S_i and first successful replan T_i,

$$\overline{T} = \frac{1}{N}\sum_{i=1}^{N}\left[\, S_i T_i + (1 - S_i)\, T_{\max}\right], \qquad T_{\max} = 30. \quad (5)$$

Equivalently, $\overline{T} = T_{\max} - \sum_{t=1}^{T_{\max}-1} F(t)$, where $F(t)$ is the fraction of episodes solved by replan t; the cumulative curves in Fig. 5 visualize F.

B.4. Deformable manipulation

Rope and granular each use one slow predictor, a DINO-WM backbone, and 50 instances revisited over four passes, with the same targets, order, and planning seeds in every pass. The slow-only variant plans from the slow predictor without fast within-episode updates. CEM uses 300 samples, 30 elites, and 10 iterations, with three MPC iterations of five pushes each. The unseen-rope evaluation freezes the predictor saved after the first pass and tests it on 91 instances absent from the learning stream, against the frozen reference on the same instances. Mean CD falls from 0.664 with the reference to 0.349 with the one-pass predictor. Two frozen granular episodes with unstable batched particle simulation were rerun individually with identical settings and seeds.

C. Baseline Checkpoints and Comparisons

Predictive world models. The comparisons focus on latent world models used for visual planning: DINO-WM, LeWM, and DDP-WM, with DINO CLS and R3M as representation baselines. Frozen and AdaJEPA use released task-specific checkpoints. For PushObj and PushT, SE-JEPA uses ResNet-based world models trained directly on task trajectories. Its PushObj backbone is trained for three epochs on 16,000 trajectories covering T, L, Z, and plus. The additional DINO-WM shape model uses the same training trajectory count and epoch budget, with 4,000 validation trajectories and batch size 32. DINO CLS and R3M use existing T-only checkpoints; LeWM and DDP-WM use their corresponding checkpoint adapters.

Offline generative baselines. For PushT we use offline-trained DreamerV3 and IRIS predictors for CEM planning on the same VIS50 targets as DINO-WM. On PushObj, Zhou et al. [42] report 18% success for DreamerV3 and 14% for IRIS (their Table 4). They train on four shapes and test on two unseen shapes, requiring both the pusher and the object to reach their targets. Table 1 uses the seven-shape benchmark in Appendix B.

Evaluation groups. Frozen (pretrained), AdaJEPA, and SE-JEPA use the paired held-out targets in Appendix B. The external visual baselines share the VIS50 targets, with 50 initial/goal pairs in each of six shifted conditions and planning seed 100. Methods within each of our evaluation groups receive the same targets and budget. PushT Shift averages blur, noise, dark, red agent, red object, and red goal; Color averages the last three. The original condition contributes to the seven-condition tables and curves.

Deformable results. The rope and granular experiments use DINO-WM with pretrained DINOv2 patch features as the frozen backbone. This complements the task-trained ResNet setting used for rigid-object manipulation. AdaJEPA adapts within each episode, whereas SE-JEPA uses a consolidated slow predictor. Italic CD entries for DreamerV3, IRIS, R3M, and DINO CLS come from the ten-instance evaluation of Zhou et al. [42]. Our frozen DINO-WM reference uses the same 50 instances as our deformable adaptation runs and appears in both the DINO-WM and Frozen rows. The repeated-instance and unseen-rope protocols are specified in Appendix B.

D. Additional Results

D.1. All object shapes

Table 6 reports every held-out shape. SE-JEPA gives the largest gains on the unseen shapes: with GD, I rises from 44.7% (AdaJEPA) to 77.6% and square from 46.7% to 61.1%. It also leads on the seen shapes in aggregate.

D.2. All visual conditions

Table 7 and Fig. 7 break the paired visual results down by condition. SE-JEPA improves on all seven conditions with both planners. The largest gains, more than 25 points, appear under red-object and red-goal recoloring, where episodic adaptation has to rediscover the new appearance in every episode.

D.3. Planning efficiency

Figure 5 plots cumulative held-out success against the number of MPC replans. SE-JEPA pulls ahead within the first four replans and keeps the lead for the rest of the budget, showing how the advantage develops before the final 30-replan evaluation.

D.4. Deformable manipulation

Figure 6 shows the four-pass learning curves in the main paper. Table 8 adds standard errors to the deformable results of Table 1. Table 9 follows consolidation pass by pass at two slow learning rates. With 10^{-4}, CD decreases at every pass on both tasks. Rope improves faster in the early passes at 2×10^{-4}; the two learning rates give similar fourth-pass CD on both tasks.

Table 5. Both planners on the paired held-out sets. Shape All and Unseen use held-out PushObj targets. Visual All averages seven conditions and Shift six; visual Steps averages all seven conditions.

		PushObj			PushT		
Planner	Method	All ↑	Unseen ↑	Steps ↓	All ↑	Shift ↑	Steps ↓
GD	Frozen	42.48	32.44	20.02	44.10	41.56	19.94
GD	AdaJEPA	72.19	55.33	14.52	58.67	54.89	17.96
GD	SE-JEPA	81.59	72.44	11.30	75.68	73.63	13.21
CEM	Frozen	43.81	36.22	21.08	50.86	48.33	19.56
CEM	AdaJEPA	70.38	54.44	16.02	62.95	60.22	17.87
CEM	SE-JEPA	79.08	69.85	13.53	77.05	76.11	14.15

Table 6. Held-out success (%) per PushObj shape. I, square, and small tee are unseen in pretraining. All averages the seven shapes.

Planner	Method	I	L	Square	Z	Small tee	Plus	T	All
CEM	Frozen	32.7	52.7	23.3	41.3	52.7	43.3	60.7	43.8
CEM	AdaJEPA	49.3	82.0	44.7	82.0	69.3	79.3	86.0	70.4
CEM	SE-JEPA	72.7	84.2	58.4	86.2	78.4	85.6	88.0	79.1
GD	Frozen	24.0	52.7	28.0	40.0	45.3	48.7	58.7	42.5
GD	AdaJEPA	44.7	90.0	46.7	82.0	74.7	82.0	85.3	72.2
GD	SE-JEPA	77.6	88.4	61.1	88.0	78.7	87.6	89.8	81.6

D.5. Component protocols and learning rates

We assign each variant the same initial checkpoint and the same sequence of shapes and goals. The comparison uses the first learning chain for direct weight inheritance and the corresponding chain for the full model. AdaJEPA starts each goal from the common checkpoint. The shared replay variant trains a single slow predictor on stored trajectories. The EMA variant uses an exponential moving average of the fast weights to update a single slow predictor.

For GD, disabling only the read protection produced identical held-out rollouts to AdaJEPA from the same checkpoint across all 21 test batches. Removing maturity screening instead allows a slow predictor to compete for retrieval earlier. This differs from the familiarity ablations in Table 2, which change how the agent handles episodes the pretrained model already predicts well. For context, a second CEM run of the full model on the same learning stream reached 80.19%, compared with 78.57% in the original run.

On PushT, doubling the slow learning rate from 10^{-4} to 2×10^{-4} changes success by $+0.19$ percentage points with GD and -1.05 with CEM. These comparisons average three learned states per planner over all seven visual conditions, including the original appearance.

D.6. Transferring fast weights across goals

We tested whether the parameter update from one episode helps on the next. Let s_e be the offset from the pretrained weights at the start of episode e, and Δ_e the fast update in that episode. We set $s_{e+1} = s_e + w\Delta_e$, where $w = 0$ resets to the pretrained model and $w = 1$ retains the full update. Each variant sees the same 100 PushObj goals in five blocks of 20 (I, square, T, I, square). Keeping the full update loses more goals than it gains relative to reset (Table 11, two-sided exact McNemar $p = 0.029$). Smaller weights give success rates closer to reset.

D.7. A seven-round shape stream

A second PushObj stream runs seven rounds with a slow learning rate of 2×10^{-4}, one chain per planner. Each round has 50 episodes per shape, drawn without replacement and excluding the held-out sources, for 2,450 episodes per planner. Online success rises from 70.6% in the first round to 78.6% in the seventh with GD, and from 72.3% to 78.3% with CEM.

The episode logs record the fast-update fingerprint of every episode. Figure 8 shows the cosine between consecutive fingerprints: the mean is 0.018 for both planners, and most of the mass lies near zero. These measurements support consolidating transition data rather than accumulating episode-specific parameter updates (Sec. 3.1).

D.8. What SE-JEPA chooses to learn

SE-JEPA learns selectively. In a seven-round PushObj stream of 2,450 episodes per planner, it writes only 25% (GD) to 27% (CEM) of episodes to memory; the rest are familiar to the pretrained model, still pending, or rejected by validation. Retrieval is just as targeted: slow predictors plan about one episode in four, and the reference handles

Table 7. Held-out visual success (%) per condition on the paired targets. All averages seven conditions; Shift excludes Original.

Planner	Method	Original	Blur	Noise	Dark	Red agent	Red object	Red goal	All	Shift
CEM	Frozen	66.0	56.0	62.7	70.7	50.0	14.0	36.7	50.9	48.3
CEM	AdaJEPA	79.3	78.0	76.7	80.0	71.3	14.0	41.3	63.0	60.2
CEM	SE-JEPA	82.7	86.4	86.7	88.0	79.6	47.1	68.9	77.0	76.1
GD	Frozen	59.3	48.7	55.3	58.7	50.7	6.7	29.3	44.1	41.6
GD	AdaJEPA	81.3	77.3	73.3	73.3	58.0	10.7	36.7	58.7	54.9
GD	SE-JEPA	88.0	88.7	84.9	87.3	74.7	44.4	61.8	75.7	73.6

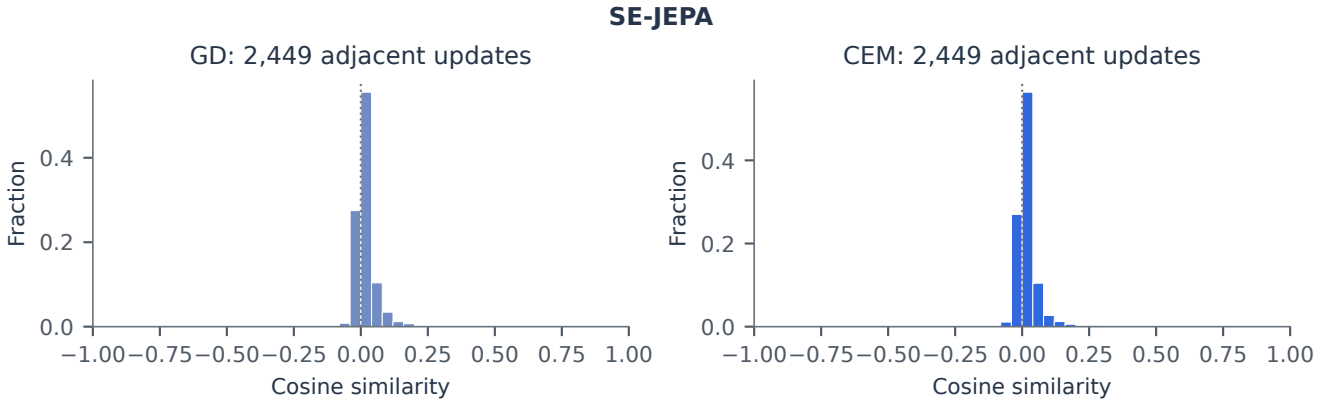


Figure 8. Cosine similarity between consecutive fast-update fingerprints in the seven-round shape stream (2,449 pairs per planner).

Table 8. Deformable manipulation: mean CD ± standard error over 50 instances. SE-JEPA uses its slow predictor after four passes.

Method	Rope CD ↓	Granular CD ↓
DINO-WM	0.491 ± 0.059	0.301 ± 0.020
LeWM	0.786 ± 0.112	0.322 ± 0.018
DDP-WM	0.677 ± 0.086	0.259 ± 0.012
AdaJEPA	0.629 ± 0.096	0.304 ± 0.019
SE-JEPA	**0.252 ± 0.045**	**0.220 ± 0.012**

Table 9. CD of SE-JEPA after each consolidation pass at two slow learning rates (mean ± standard error, 50 instances).

	lr 10^{-4}		lr 2×10^{-4}	
Pass	Rope	Granular	Rope	Granular
1	0.469 ± 0.100	0.257 ± 0.014	0.416 ± 0.070	0.260 ± 0.012
2	0.342 ± 0.054	0.235 ± 0.012	0.286 ± 0.050	0.241 ± 0.012
3	0.282 ± 0.033	0.224 ± 0.010	0.283 ± 0.037	0.247 ± 0.012
4	0.252 ± 0.045	0.220 ± 0.012	0.243 ± 0.033	0.243 ± 0.013

Table 10. Success on all seven held-out PushObj shapes and mean replans as a function of the slow learning rate. All runs use GD and the same stream as the component comparison in Table 3.

Slow learning rate	Success (%)	Replans
10^{-4} (default)	81.43	11.37
5×10^{-5}	79.62	11.79
2×10^{-4}	82.38	11.26
3×10^{-4}	80.57	11.72
5×10^{-4}	78.38	12.40
10^{-2}	76.95	12.54

Table 11. Fraction of each fast update retained between episodes. Wins and losses count goals reached only by that variant and only by episodic reset, respectively.

Accumulation w	Success (%) ↑	Wins / losses
0 (episodic reset)	62.0	—
0.003	64.0	7 / 5
0.01	59.0	5 / 8
0.03	54.0	4 / 12
1 (keep all)	48.0	11 / 25

the rest.

Selective writing leaves familiar dynamics to the pretrained model, which SE-JEPA never overwrites. On the four pretraining shapes, SE-JEPA reaches 86.0% held-out success with CEM and 88.4% with GD, against 82.3% and 84.8% for AdaJEPA. This preserves a useful pretrained reference while allowing specialized predictors to add capacity for recurring shifts.

E. Qualitative Examples and Reproducibility

E.1. Full trajectories

Figure 3 selects an unseen-I and a red-object episode where both baselines fail and SE-JEPA succeeds under GD. These illustrate changes in planning behavior; aggregate success is evaluated on the complete held-out sets. The source package supplies the first held-out target of test seed 2000 for each condition, and the displayed conditions are chosen by outcome. All methods share identical initial and goal observations. The panels use a consistent crop size to enlarge the movable objects, with every colored foreground element preserved.

Figures 9 and 10 provide additional CEM examples on the first held-out target of the same test seed: an unseen square and a red-agent shift. They show the complete trajectories and each method's prediction loss. These are separate from the GD examples in the main paper.

Each loss curve is measured in the latent space of its own model, so it describes how prediction error evolves within a method over the episode. Task success, judged in the shared simulator state, is the common yardstick across methods.

E.2. Reproducibility

Tables and plots use the bundled reporting snapshot and the archived Drive experiment packages. The revised trajectory panels are rendered directly from the packages' simulator frames; the asset manifest records their source files and hashes. Success and Steps are computed from the final episode files and their run configurations. Offline world-model checkpoints record the upstream code revision, the epoch budget, the model configuration, and hashes of the adapter sources. Code, configurations, and learned memory states will be released.

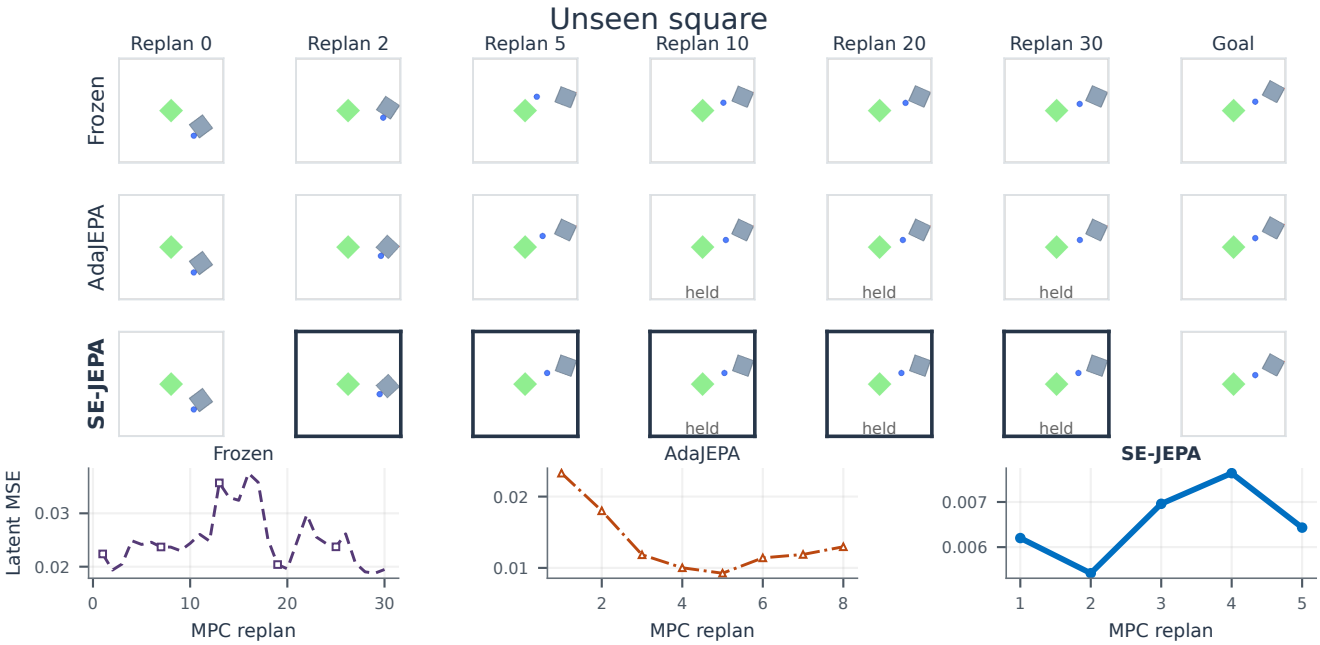


Figure 9. Full unseen-square example with intermediate frames and prediction-loss traces. Frames marked held repeat the terminal state after success. **SE-JEPA** reaches the goal at replan five, AdaJEPA at replan eight, and the frozen model does not succeed within the budget.



Figure 10. Full red-agent example, replayed in the original simulator from the executed actions. **SE-JEPA** succeeds at replan ten; Frozen and AdaJEPA reach the 30-replan limit.